Enhancing Optical characteristics of Ti_3C_2 MXene QDs through hydrothermal and electrochemical synthesis methods

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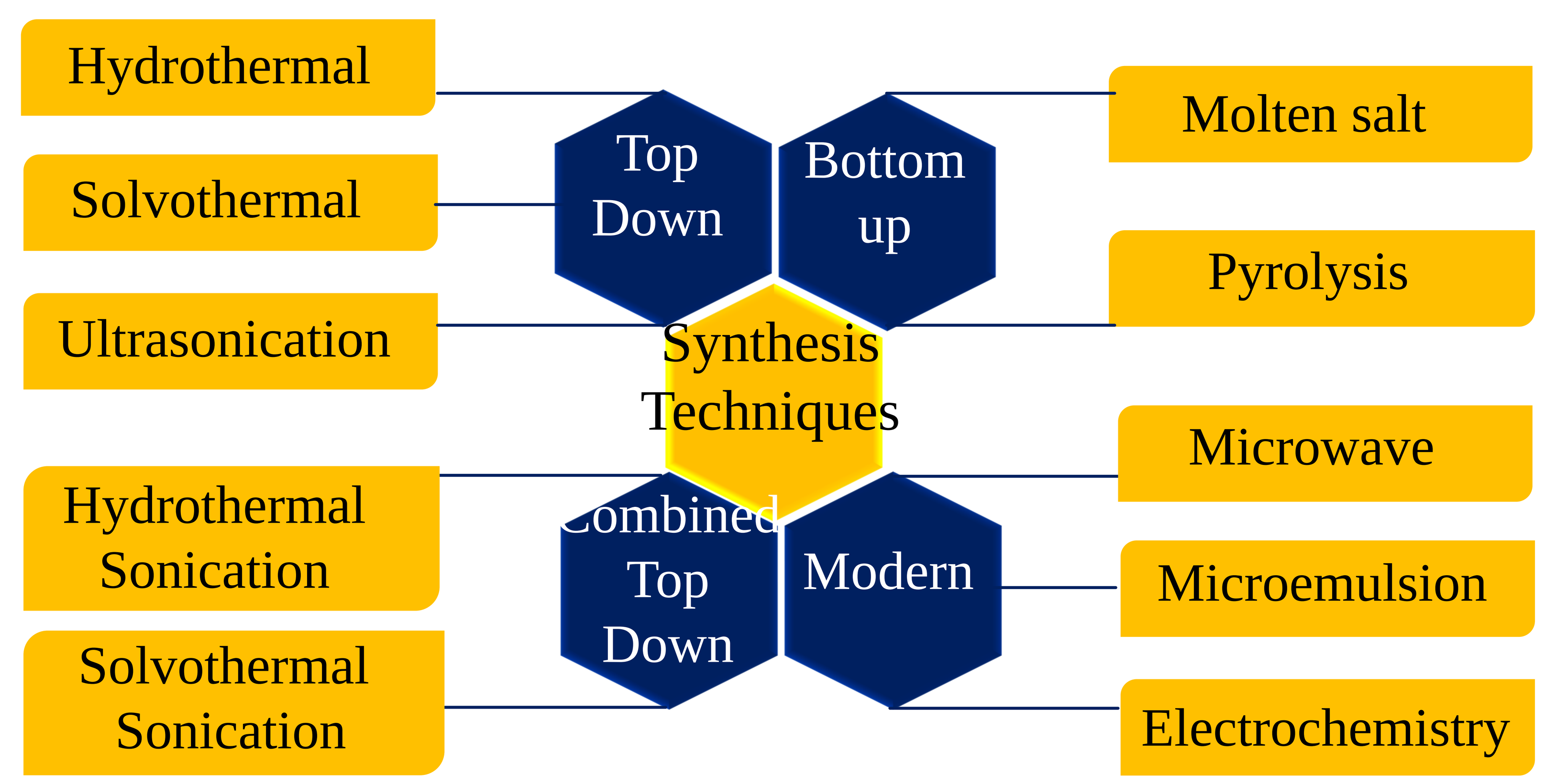
Abstract

In this study, Ti_3C_2 QDs were synthesized via two distinct methodologies: hydrothermal and electrochemical routes. The hydrothermal approach involved the use of LiF and HCl to etch titanium aluminum carbide (Ti_3AlC_2), followed by controlled heating of the decanted solution to produce the QDs. Conversely, the electrochemical method employed a three-electrode configuration with Ti_3AlC_2 as the working electrode and an ionic liquid electrolyte to synthesize the QDs directly. Atomic force microscopy (AFM) was used to measure the size of the QDs, which fell within the nanometer range, confirming the formation of QDs. Optical properties were assessed through UV-Vis spectroscopy and fluorescence lifetime measurements, showing distinct absorption and emission profiles for both sets of QDs. The hydrothermally synthesized QDs exhibited UV absorption between 250-290 nm and visible emission from 320-380 nm. Conversely, electrochemically synthesized QDs absorbed violet to blue light 320-380 nm with broader emission, indicating the potential for various fluorescence imaging applications. This study showcases an expandable strategy for creating high-grade QDs with tunable optical features, catering to the needs of advanced optoelectronic devices.

Introduction

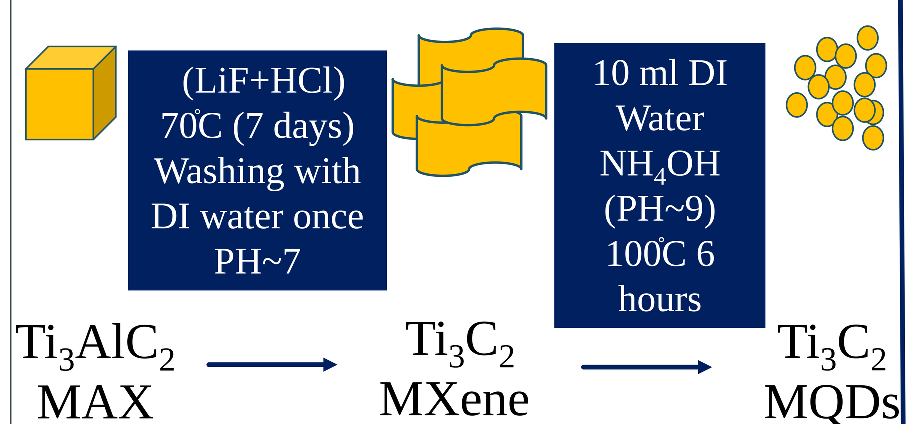
Applications

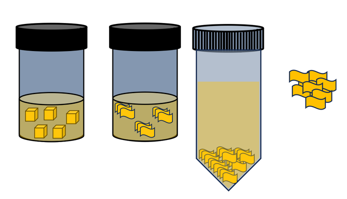
- Energy conversion and storage
- Detection of Heavy Metals
- Sensing of Biomolecules
- Biomedical Application

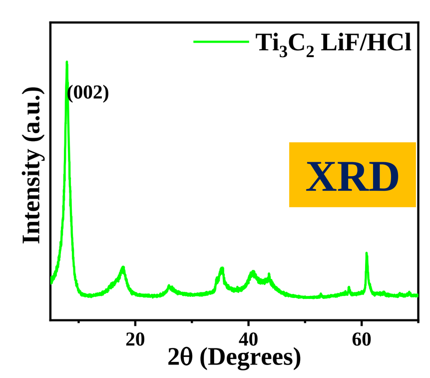


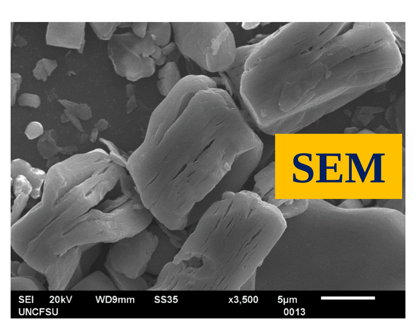
Synthesis and characterizations of Ti_3C_2 MQDs

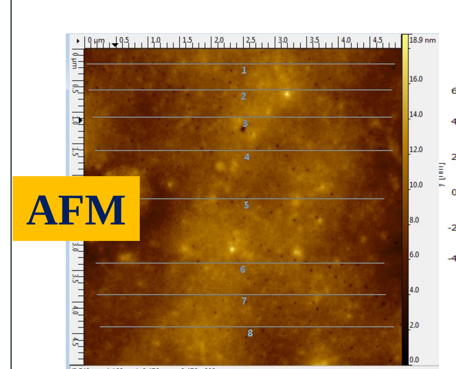
Top Down method: Hydrothermal

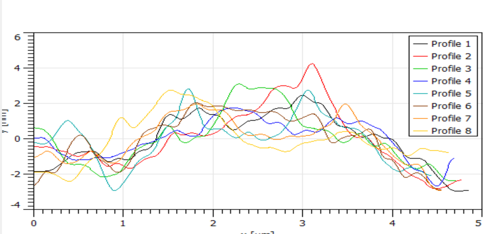




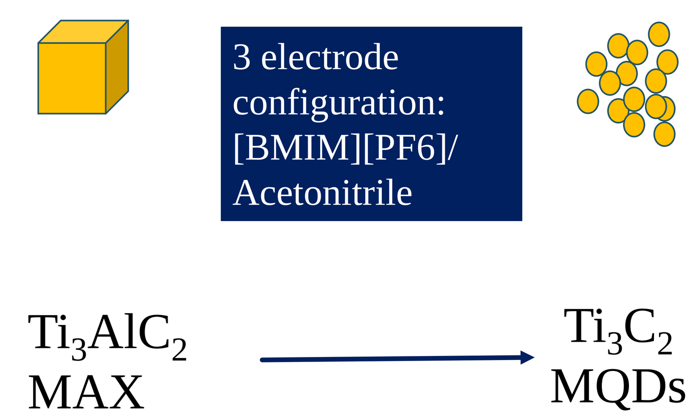






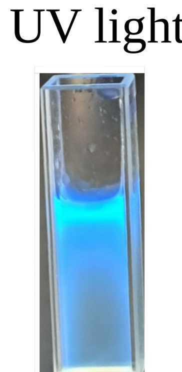


Modern method: Electrochemical

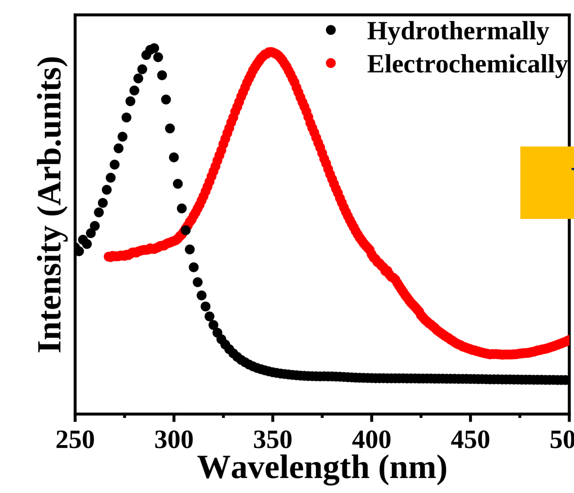


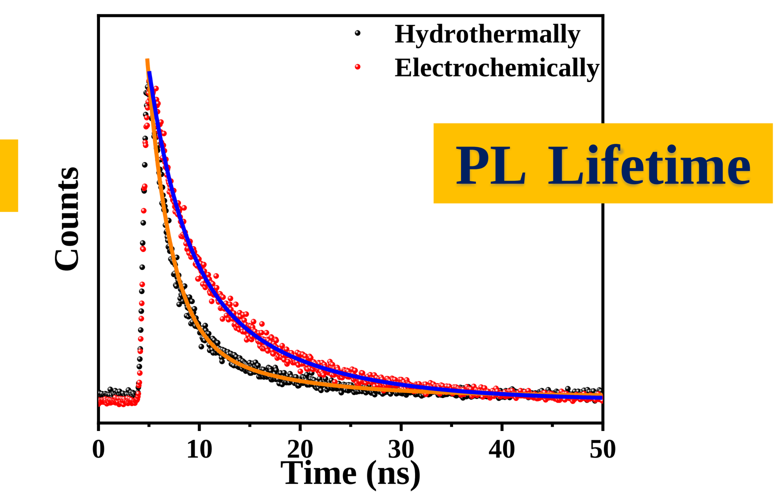
Fluorescence under UV light

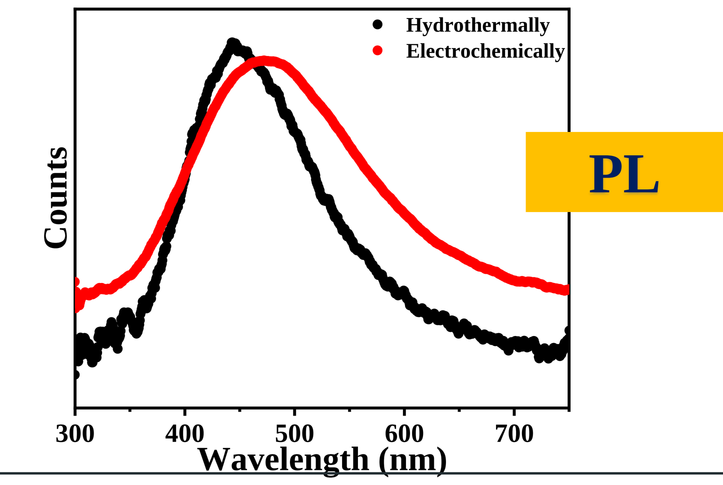
Hydrothermal



Electrochemical







- UV-Vis absorbance
  - Hydrothermally : 289 nm.
  - Electrochemically : 350 nm.
- Lifetime measurements
  - Hydrothermally : 9.82 ns
  - Electrochemically :10.9 ns

Summary and Future Work

- Electrochemically synthesized MQDs can be produced directly from the MAX phase.
- Hydrothermal and Electrochemically synthesized MQDs shows change in optical features.
- In Future, Optimize the synthesis technique for the Electrochemically synthesized MQDs.
- Study Structural and optical characterization of MQDs
- MQDs can be utilized for the Electrochemical sensors to detect organophosphate pesticides

Acknowledgements

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